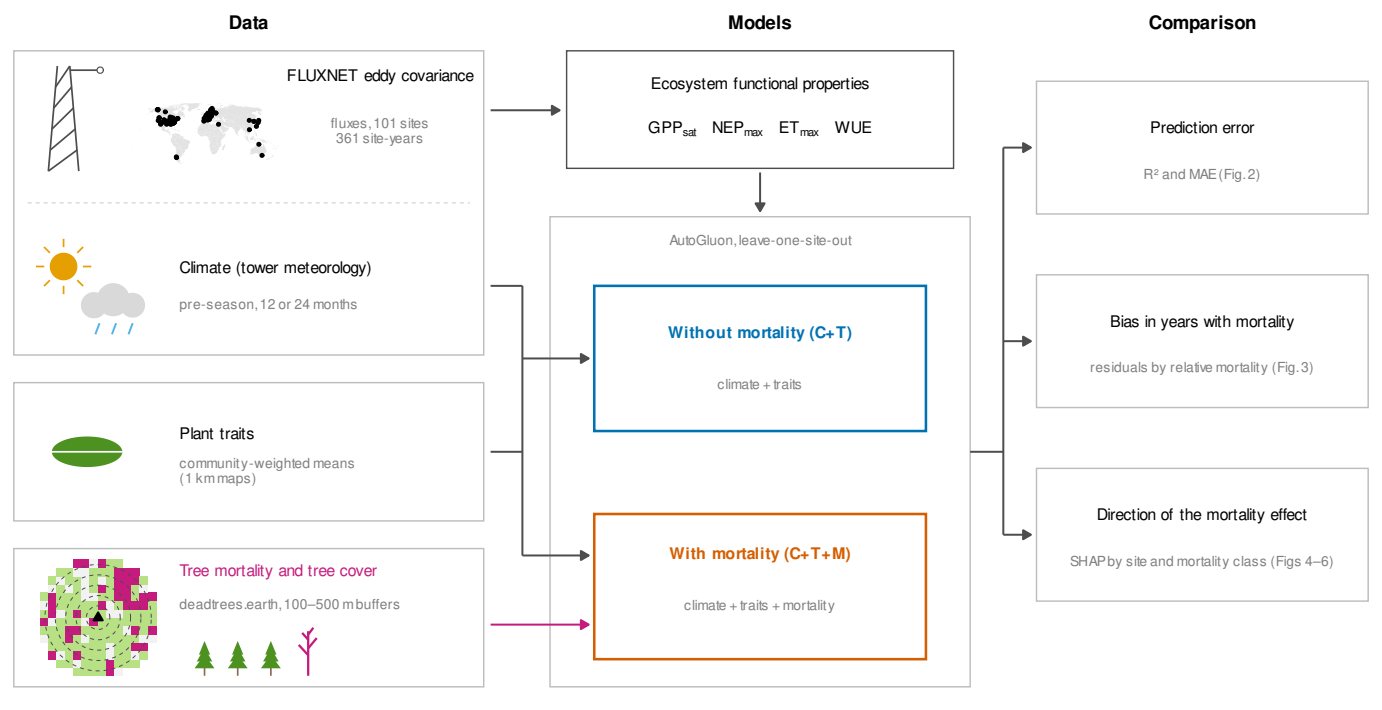

Data
Models
Comparison
FLUXNET eddy covariance
fluxes, 101 sites
361 site-years
Climate (tower meteorology)
pre-season, 12 or 24 months
Plant traits
community-weighted means
(1 km maps)
Tree mortality and tree cover
deadtrees.earth, 100–500 m buffers
Ecosystem functional properties
GPP$_{sat}$ NEP$_{max}$ ET$_{max}$ WUE
AutoGluon, leave-one-site-out
Without mortality (C+T)
climate + traits
With mortality (C+T+M)
climate + traits + mortality
Prediction error
R² and MAE (Fig. 2)
Bias in years with mortality
residuals by relative mortality (Fig. 3)
Direction of the mortality effect
SHAP by site and mortality class (Figs 4–6)